

< Prev

Aptitude ends in 27 min

20. Rob and his old calendars

Rob is looking through some old calendars and finds that the 21st of August 1995 was a Monday.

If 1996 was a leap-year, what day of the week was the 24th of August 1998?

Aptitude

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 

Analog

Starts in 27 min

[Go to Analog >](#)

Digital

Answer Options

Select any one option

☒ Monday 

☐ Thursday

☐ Friday

☐ Sunday



2. A herbal energy drink

Argument: A specific brand's herbal energy drink was given to a group of 50 consumers and a study was conducted to prove the benefits of the drink on the mood of a person. Consumers were put in time bound situations that and then given a glass of the drink to help them focus. The test results were out and 80 percent of the consumers agreed the energy drink in fact did help with their productivity. Hence, it can be proven that the specific brand's energy drink on which the survey was conducted is the best energy drink in the market.

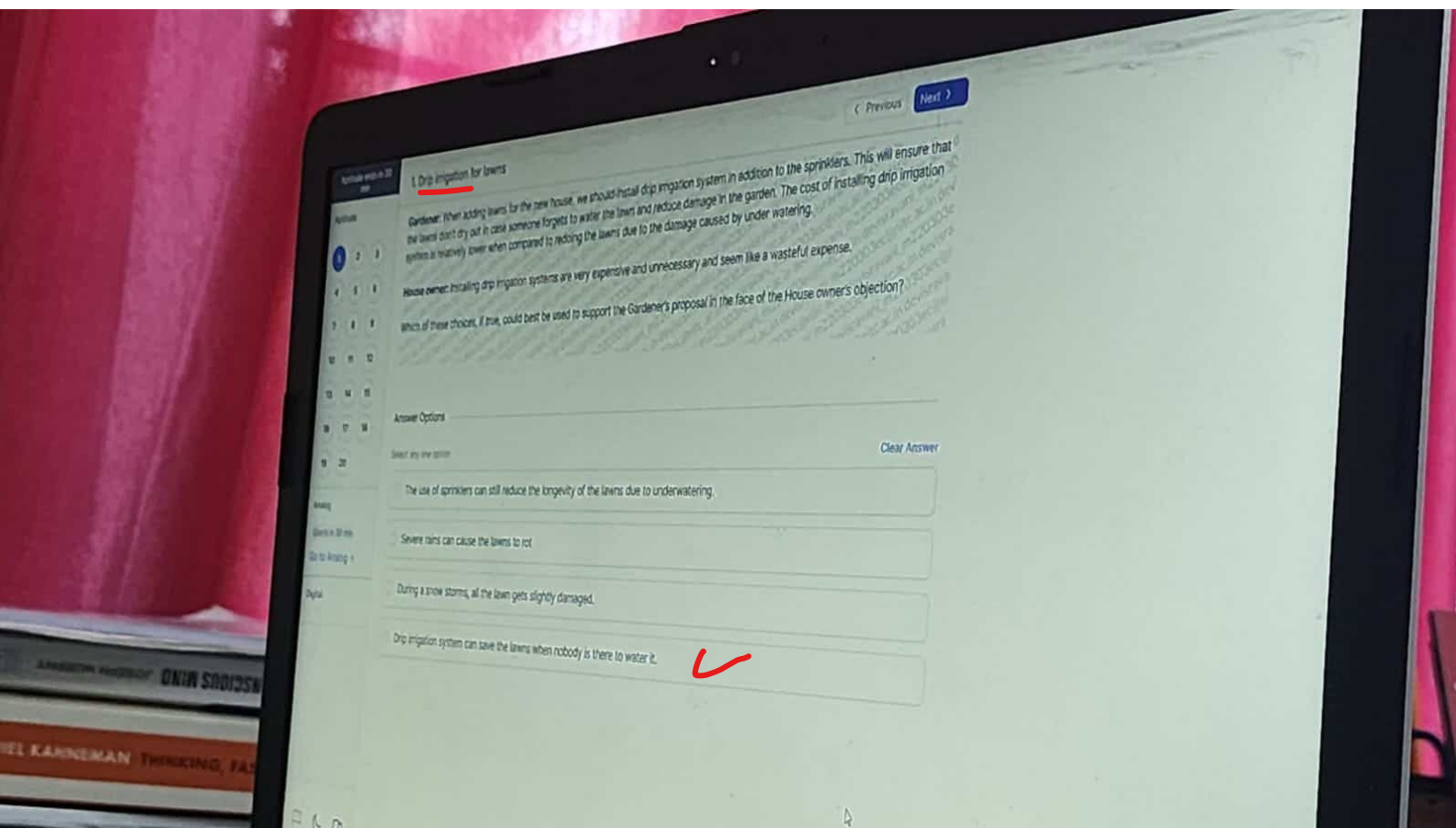
The above given argument's reasoning is most vulnerable to criticism on the grounds that it fails to consider which of the given choices?

Answer Options

Select any one option

Clear Answer

- ☐ Any of the consumers that participated in the survey are allergic to some herbs in the drink.
- ☐ The herbs in the energy drink is not very harmful to the health of the consumer.
- ☐ The price of the energy drink when compared to other brands is high.
- ☒ Customers had the opportunity to taste the energy drink from various other brands as well



3. Jia's pocket money

Jia receives an amount X as pocket money every month. Of the total amount received, Jia spends 20% on necessary purchases and 5% of the remaining amount on transportation.

If she is left with Rs. 1520, the what is X in the given context?

Answer Options

Select any one option

Clear Answer

☒ Rs. 2000

☐ Rs. 5500

☐ Rs. 2500

☐ Rs. 6080

Aptitude ends in 29 min

Aptitude

- 1
- 2
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- 19
- 20

Analog

Starts in 29 min

Go to Analog >

Digital

5. Profits made by Mrs. Lindsay

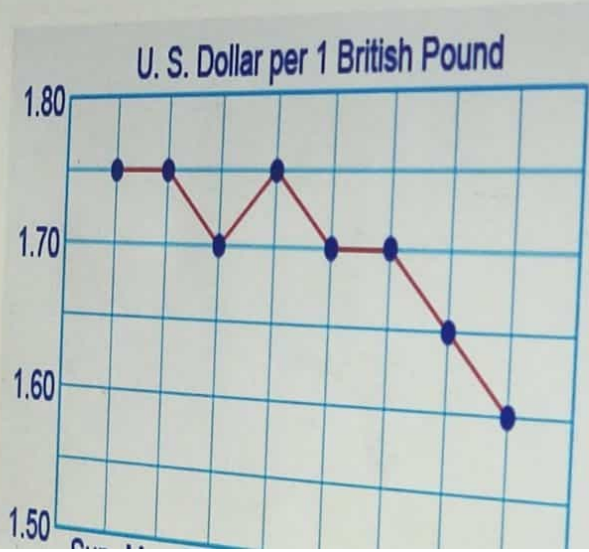
The given line graphs show the exchange rate dynamics during one week.

On Tuesday, Mrs. Lindsay bought 200 British Pounds using US Dollars. On Wednesday, she exchanged these 200 British Pounds back into US Dollars. What profit did Mrs. Lindsay make?

Note: Assume that there was zero transaction cost.

6.25

10



Aptitude ends in 29 min

4. A hexagonal table

Six friends Adam, Abe, Pepa, Jill, Bob and Ron are sitting around a hexagonal table each at one corner facing the centre. Adam is second to the left of Ron. Abe is neighbour of Pepa and Jill. Bob is second to the left of Jill. Who is sitting to the opposite of Adam?

Answer Options

Select any one option

[Clear Answer](#)

☐ Bob

☒ Jill

☐ Ron

☐ Pepa

Aptitude

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Analog

Starts in 28 min

[Go to Analog >](#)

Digital

< Previous

Aptitude ends in 30 min

Aptitude

- 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

3. Jia's pocket money

Jia receives an amount X as pocket money every month. Of the total amount received, Jia spends 20% on necessary purchases and 5% of the remaining amount on transportation.

If she is left with Rs. 1520, the what is X in the given context?

Answer Options

Select any one option

Clear Answer

☒ Rs. 2000

☐ Rs. 5500

☐ Rs. 2500

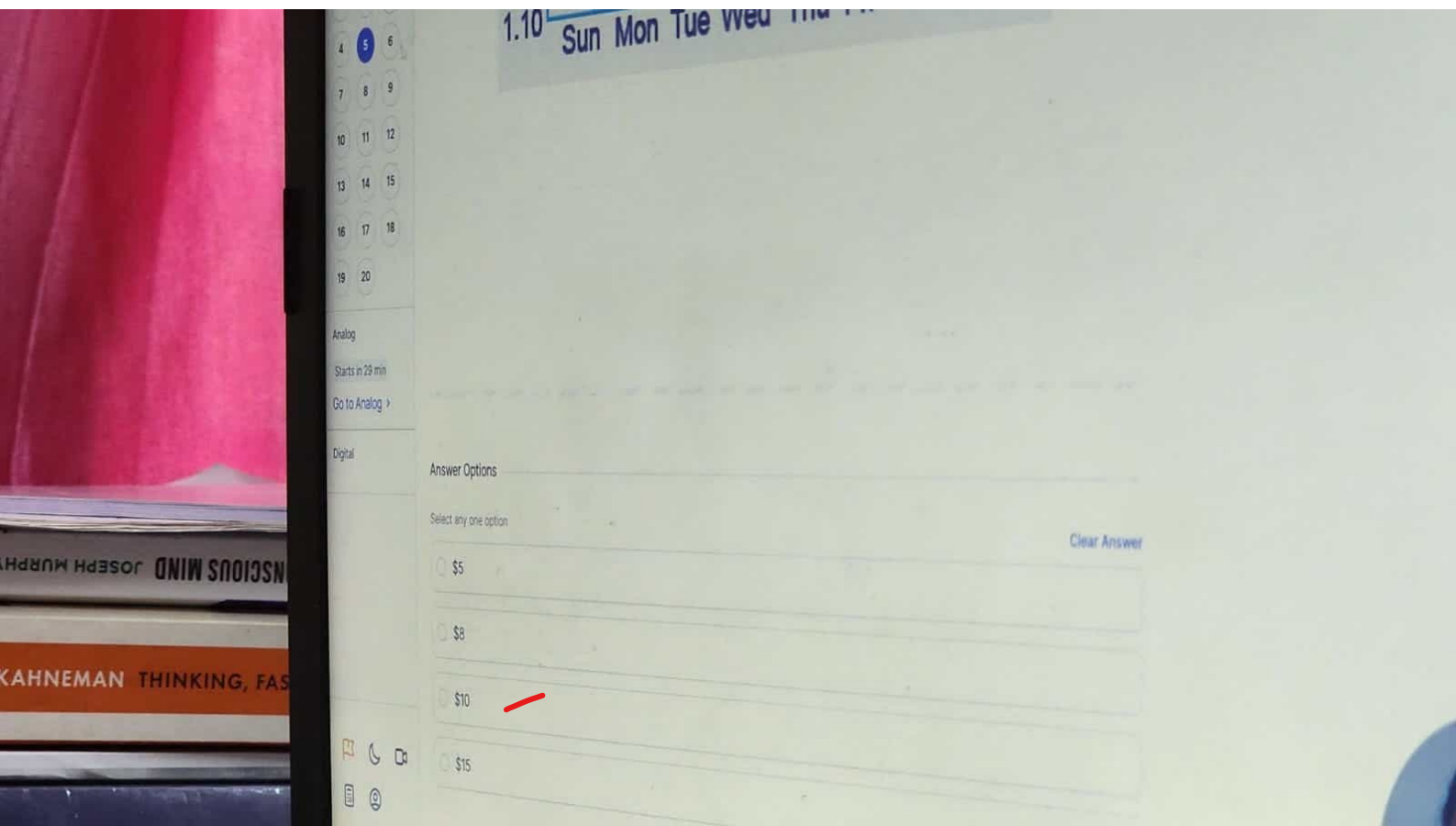
☐ Rs. 6080

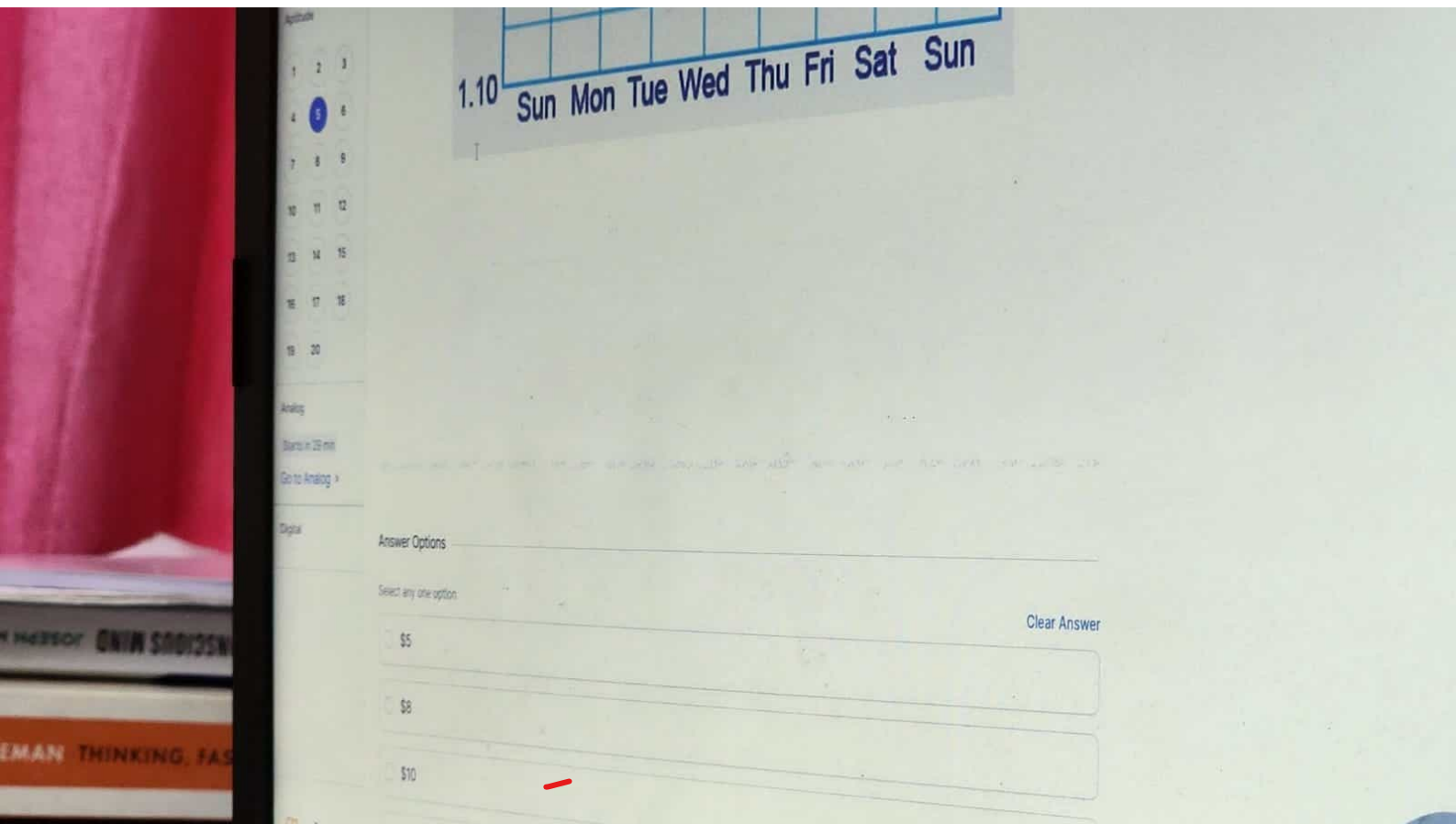
Analog

Starts in 30 min

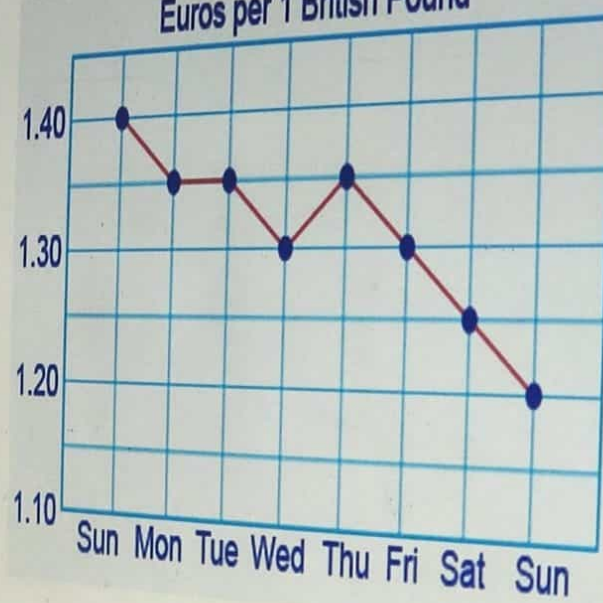
Go to Analog >

Digital





Euros per 1 British Pound



Apptitude ends in 29 min

Apptitude

- 1
- 2
- 3
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- 5
- 6
- 7
- 8
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- 10
- 11
- 12
- 13
- 14
- 15
- 16
- 17
- 18
- 19
- 20

Analog

Starts in 29 min

Go to Analog

6. A mocktail

Bob is preparing a mocktail for his friends by mixing 10 parts water and 5 parts of a mocktail mix in a large jug.

How much of the mixture should he draw out of the jug and replace it with the mocktail mix to ensure that the mixture is half water and half mocktail mix?

Note: Assume that the jar contains 15 litres of the mixture.

$$\frac{3.75}{15}$$

Answer Options

Select any one option

☒ 1/4

☐ 1/8

☐ 1/2

Clear Answer

6. A mocktail

Bob is preparing a mocktail for his friends by mixing 10 parts water and 5 parts of a mocktail mix in a large jug.

How much of the mixture should he draw out of the jug and replace it with the mocktail mix to ensure that the mixture is half water and half mocktail mix?

Note: Assume that the jar contains 15 litres of the mixture.

Answer Options

Select any one option

☐ 1/4

☐ 1/8

☐ 1/2

Clear Answer

- 7
- 8
- 9
- 10
- 11
- 12
- 13
- 14
- 15
- 16
- 17
- 18
- 19
- 20

Note: Assume that the pe

Answer Options

Select any one option

☒ $1/4$

☐ $1/8$

☐ $1/2$

☐ $1/3$

Clear Answer

Analog

Starts in 29 min

Go to Analog >

Digital

SCIOUS MIND JOSEPH MUR

INEMAN THINKING, FA

Aptitude ends in 29 min

Aptitude

1 2 3

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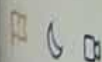
19 20

Analog

Starts in 29 min

Go to Analog >

Digital



Submit test

< Previous

Next >

8. A rectangular plot

A rectangular plot has been allotted by the government for the construction of an embassy building. The length of the plot is increased by 10% to equip a cafeteria.

By what percent should the breadth of the plot be decreased such that the area of the plot is same as before?

Answer Options

Select any one option

Clear Answer

☐ 120/13

☐ 100/11

☐ 120/11

☐ 100/9

< Previous

Next >

Aptitude ends in 29 min

7. Bob's shop

Bob is a retailer selling rice in his shop. He mixes two varieties of rice R1 and R2 together in the ratio 4:7 and sells this mixture at a price of Rs. 17 per kg. Due to this, he incurs a loss of 15%.

In the given context, what is the cost price per kg of R1 if the cost price of R2 is Rs. 22 per kg?

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Answer Options

Select any one option

Clear Answer

☐ Rs. 15

☒ Rs. 16.5

☐ Rs. 17

☐ Rs. 17.5

Analog

Starts in 29 min

Go to Analog >

Digital



Submit test

Aptitude ends in 28 min

Aptitude

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Analog

Starts in 28 min

Go to Analog >

Digital

9. Five students

Five students are made to stand in a row as given in the following statements.

Statements:

1. A and B are standing next to each other.
2. C and D are standing next to each other.
3. A is not standing next to E who is standing at the left end of the line.
4. B is standing in the second position from the right.
5. C is to the right of both D and E.
6. B and C are standing together.

Which of the given choices is/are **TRUE** regarding the following conclusions made from the given statements?

Conclusions:

1. B is placed between C and A.
2. A is placed between C and B

E

Answer Options

Select any one option.

☐ Only (1) conclusion is incorrect

☒ Only (2) conclusion is incorrect

☐ Either (1) or (2) is incorrect

☐ Neither (1) nor (2) is incorrect



Submit test

11. Grids A and B

The two grids A and B given below follow a certain pattern. Analyse A and determine the correct value of '?' in B.

3	9	4
8	2	14
3	12	6

A

4	13	5
?	4	15
2	12	6

B

Answer Options

Select any one option

10

11

12

14

Clear An

< Previous

Next >

Time ends in 28 min

10. Bella arranging books

Bella is arranging her books on a new bookshelf. She has placed B1 in the 24th position from the left and B2 in the 27th position from the right.

What is the minimum number of books that Bella owns if only B3 is exactly in between B1 and B2?

Note: Bella is placing all her books on the bookshelf.

Answer Options

Select any one option

Clear Answer

☐ 58

☐ 55

☒ 48

☐ 50

Time ends in 28 min

Analog >



Submit test

< Previous

Next >

12. Cats and dogs

Given below are 4 statements followed by 4 conclusions numbered 1 to 4. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. Choose the most appropriate option for the given conclusions that logically follows from the given statements, disregarding commonly known facts.

Statements:

Some cats are dogs. No dog is a frog. All frogs are bats. Some bats are eagles.

Conclusions:

1. Some frogs are not eagles.
2. No dog is a bat.
3. Some cats are bats.
4. No eagle is a cat.

SR

Answer Options

Select any one option

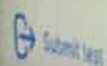
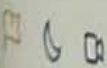
Clear Answer

☐ Both conclusion 2 & 4 follow

☐ Conclusions 2, 3 & 4 follow

☐ Only conclusion 1 follows

☐ No conclusion follows



Aptitude ends in 28 min

Aptitude

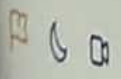
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- 20

Analogy

Starts in 28 min

Go to Analogy >

Digital



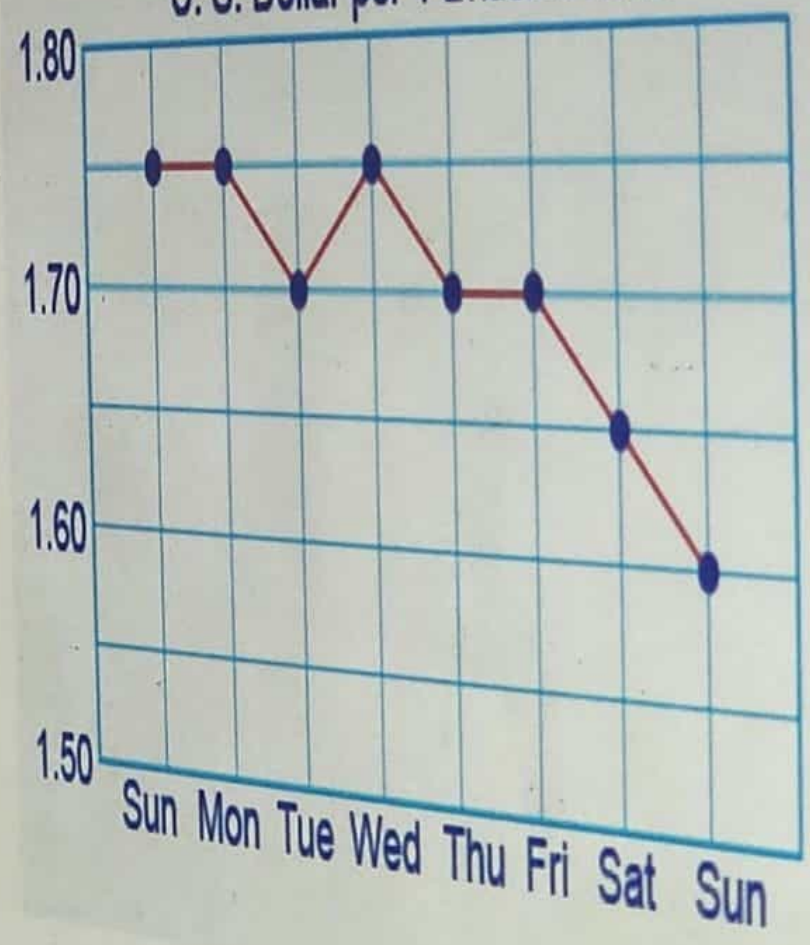
Submit test

14. The exchange rate dynamics

The given line graphs show the exchange rate dynamics during one week.

On the basis of the line graphs, how much did the dollar depreciate against the euro during this seven-day period?

U. S. Dollar per 1 British Pound



Euros per 1 British Pound



Aptitude ends in 28 min

13. Finding value of expression

In a certain system, two positive integers are subtracted in a certain way. Given below are some examples:

1. $62 - 56 = 0$

2. $48 - 32 = -1$

3. $55 - 31 = 4$

Analyse the given expressions and determine the result of the expression **42 - 28**.

Aptitude

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Answer Options

Select any one option

☒ 1

☐ 2

☐ 3

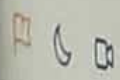
☐ 4

Analog

Starts in 28 min

[Go to Analog >](#)

Digital



[Submit test](#)

< Previous

Next >

15. A chess club

A chess club has three teams T1, T2 and T3 out of which - one team always tells the truth, one team always lies and one alternates between telling the truth and lying. The teams get together only thrice a week, i.e., Tuesday, Friday and Sunday. One of these days, the three teams make the following statements:

T1: "Tuesday is the most comfortable day to get together and Friday is comfortable for T2".

T2: "Friday is comfortable for T3 and Sunday is comfortable for T1".

T3: "Tuesday is the most comfortable day to get together and T1 is comfortable with Sunday".

If all teams prefer a different day to get together, then which team is telling the truth?

Note: All teams works as a single union.

Answer Options

Select any one option:

Clear Answer

☒ T1

☐ T2

☐ T3

☐ Cannot be determined

Apitude ends in 28 min

14. The exchange rate of

Apitude

- 1 2 3
- 4 5 6
- 7 8 9
- 10 11 12
- 13 14 15
- 16 17 18
- 19 20

Analog

Starts in 28 min

Go to Analog >

Digital



Answer Options

Select any one option

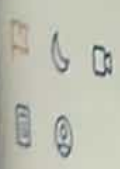
☐ 5.00%

☐ 5.50%

☒ 6.25%

☐ 7.50%

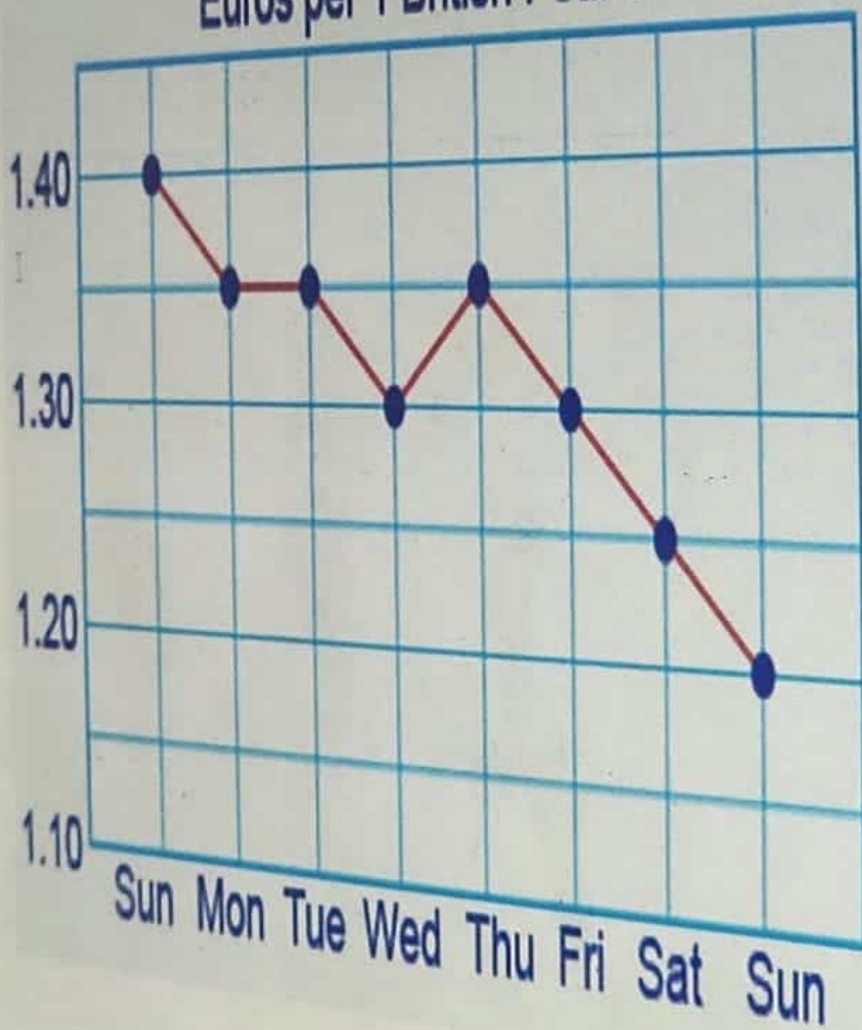
Clear



Submit test



Euros per 1 British Pound



17. Identify the triangles

17

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Answer

Start in 27 min

Go to Analog >

Original

Answer Options

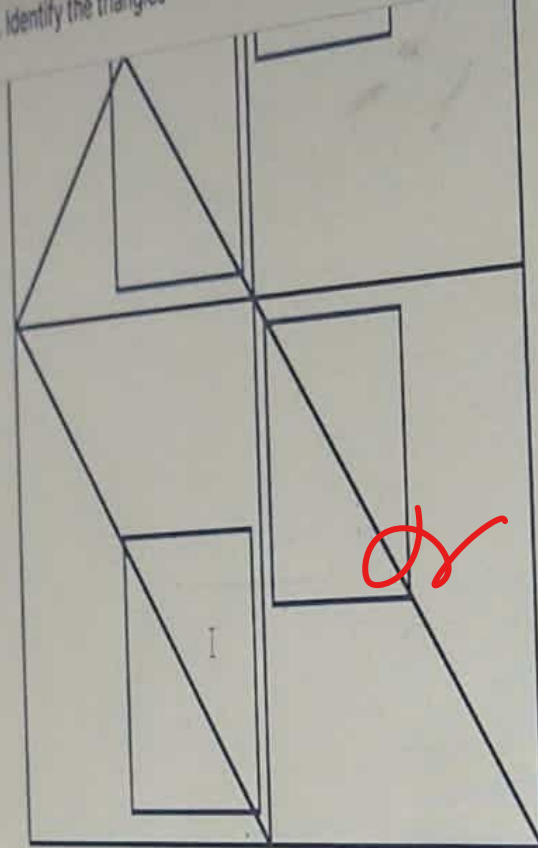
Select any one option

☐ 21

☐ 20

☐ 18

☐ 19

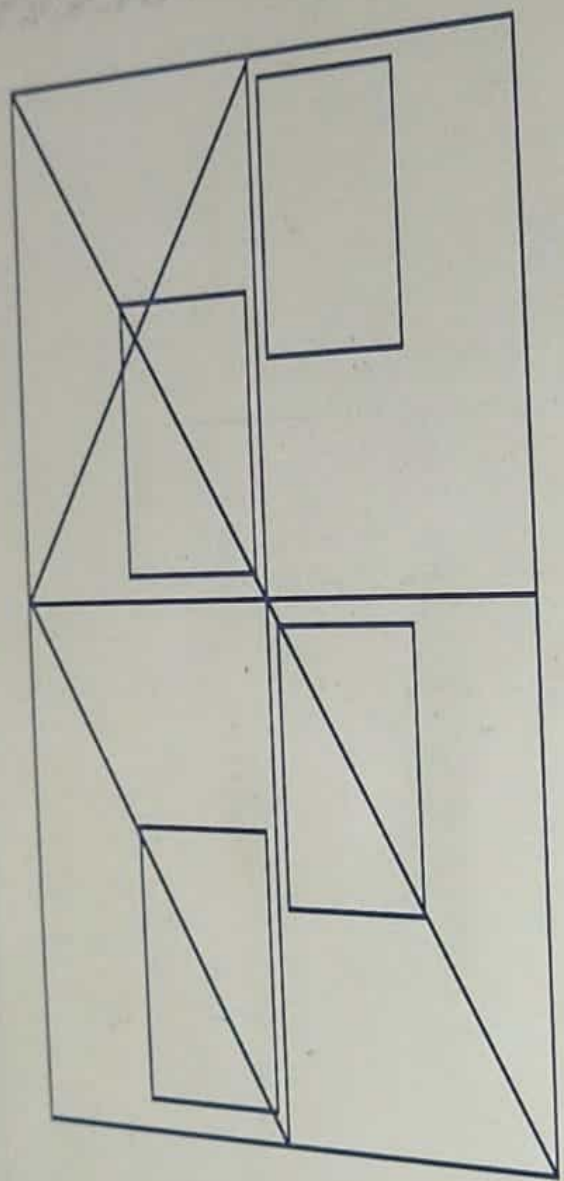


Applicable ends in 27 min

17. Identify the triangles

How many triangles can be found in the image given below?

- 1 2 3
- 4 5 6
- 7 8 9
- 10 11 12
- 13 14 15
- 16 17 18
- 19 20

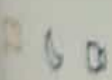


Answer Options

Select any one option

21

20



< Previous

Next >

16. A primary school teacher

A primary school teacher wants to arrange a visit to the museum which is 600 miles away from the school. She arranges a school bus for the same and ensures that the bus has a full tank. However, the bus driver does not remember the capacity of the gas tank nor does he remember the efficiency of the engine.

Which of these statements would be sufficient to determine if the bus has enough gas to reach the museum?

Statement 1: The driver is driving at a constant speed.

Statement 2: The driver realizes that he has three-quarters of the full tank of gas left after he drives 150 miles.

Statement 3: The driver realizes that he has three-quarters of the full tank of gas left after he drives 450 miles.

Answer Options

Select any one option

Clear Answer

☐ Both Statements 1 and 2 are sufficient

☐ Both Statements 1 and 3 are sufficient

☐ All three Statements are sufficient

☒ Statements 1 and 2 OR Statements 1 and 3 is sufficient

☐ None of the Statements are sufficient

1

2

3 $\rightarrow A$

4 $\rightarrow D$

5 $\rightarrow C$

6 $\rightarrow D$

7 $\rightarrow B$

8 $\rightarrow B$

9 $\rightarrow B$ [only 2 is incorrect]

10

11 $\rightarrow A$

12 $\rightarrow$ only conclusion 1 follows

13 $\rightarrow C$

14

15

16 $\rightarrow A$

17 $\rightarrow A$

~~18~~ $\rightarrow B$

~~19~~ $\rightarrow D$

20 $\rightarrow A$

< Previous

Next >

19. Selecting fudge balls

A dessert shop sells assorted boxes of chocolates every holiday season. Each of these boxes contains four caramel twists, three dark chocolate truffles and two fudge balls. John buys a box and successively picks three pieces of chocolates from it.

In the given context, what is the probability that John selects **exactly one fudge ball** from the box?

Answer Options

Select any one option

Clear Ans

☒ 1/2

☐ 2/3

☐ 3/4

☐ 1/4

< Previous

Next >

18. A luxury perfume

A luxury perfume launched its perfumes on March 8, 1997 with a stock of x bottles of perfume. By Dec 31 of that same year, $\frac{2}{7}$ of the x bottles are sold. This pattern continues such that by the end of each subsequent year $\frac{2}{7}$ of the stock that was there at the beginning of that year is sold out.

Analyse the given scenario and determine the year in which the stock of these bottles is reduced to less than $\frac{1}{4}$ of the original x bottles.

Note: Assume that only one type of perfume is being sold and the bottles are not restocked.

SB

Answer Options

Select any one option

Clear Answer

☐ 2003

☐ 2002

☐ 2001

☐ 2000

Analog ends in 44 min

Apptitude

Submitted

Analog

- 1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

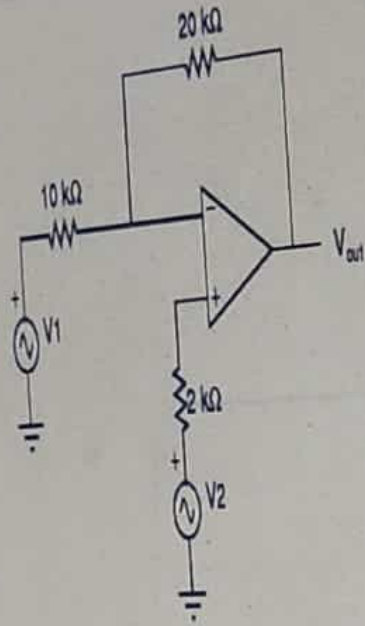
Digital

Starts in 44 min

Go to Digital >

7. Output of the opamp

Consider the circuit shown in the figure. What is the output of the op amp?



Answer Options

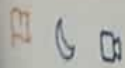
Select any one option

☐ $2(V_2 - V_1)$

☐ $(3V_2 - 2V_1)$

☐ $(2V_1 - 3V_2)$

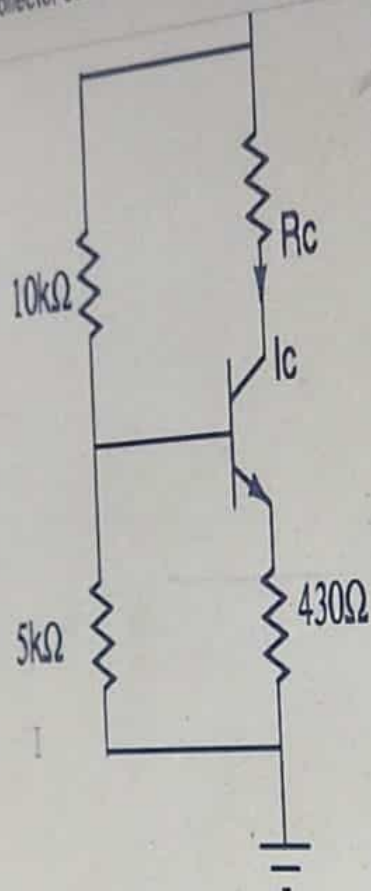
☐ $(3V_1 - 2V_2)$



Submit test

Analogy ends in 43 min

11. Collector current of a transistor



Digital

Starts in 43 min

Go to Digital >

Answer Options

Select any one option

☐ 1 mA

☐ 5 mA

☐ 10 mA

☐ Can not be determined since R_c is not given

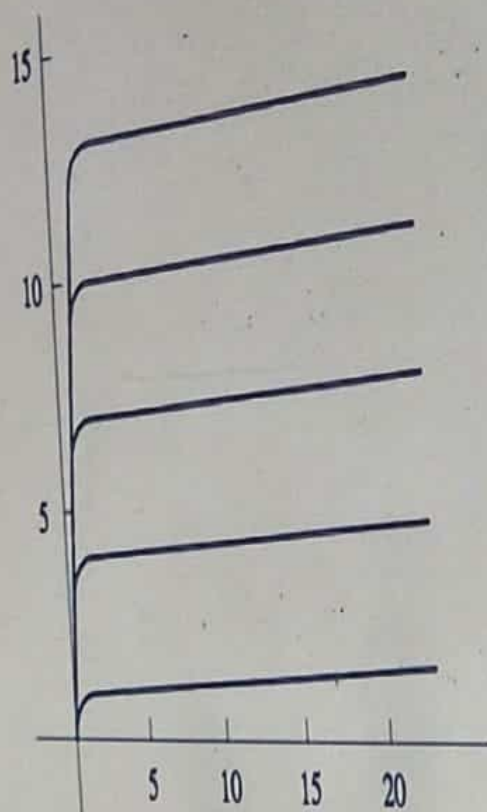
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< Previous

Next

17. Characteristics of a BJT

Which of these choices is **incorrect** about the output characteristics of a BJT operating in common-emitter configuration?



Answer Options

Select any one option

- ☐ The graph has collector-emitter voltage along x-axis.
- ☐ The graph has collector current along the y-axis.
- ☐ Each curve in the graph corresponds to a different value of base current.
- ☐ None of these

< Previous

Next >

14. Using a 10-bit A/D converter

You are using a 10-bit analog-to-digital converter with reference voltage of 5V to digitize an analog signal in the 0 to 5 V range.

What is the maximum peak-to-peak ripple voltage that can be allowed in the reference voltage?

Answer Options

Select any one option

☐ nearly 100 mV

☐ nearly 50 mV

☐ nearly 25 mV

☐ nearly 5 mV

Clear

Starts in 43 min

2 3

5 6

8 9

11 12

14 15

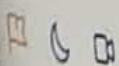
16 17 18

19 20

Digital

Starts in 43 min

Go to Digital >



Submit test

Analog ends in 44 min

10. A series RLC circuit

A series RLC circuit has a resonance frequency of 40 MHz.

Which of the given deductions can be made from the above information?

Amplitude

Submitted

Analog

1

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11

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Digital

Starts in 44 min

Go to Digital >

Answer Options

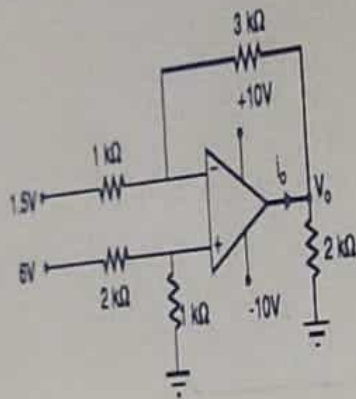
Select any one option

- ☐ At 40 MHz, the circuit shows a resistive impedance of R
- ☐ For frequencies lower than 40 MHz, the circuit behaves like an inductive circuit
- ☐ For frequencies greater than 40 MHz, the circuit behaves like a capacitive circuit
- ☐ All of the above

Analogue ends in 43 min

16. An ideal opamp circuit

Find the value of the output voltage V_o for the given ideal op amp circuit.



Answer Options

Select any one option

☐ 3.5V

☐ 4.0V

☐ 4.5V

☐ 3.0V

Analog ends in 43 min

Aptitude

Submitted

Analog

- 1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

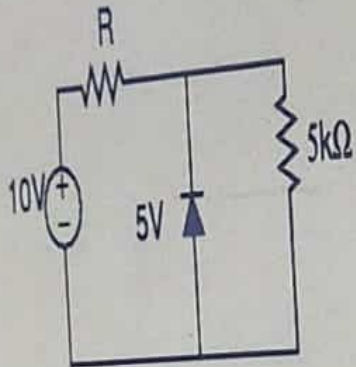
Starts in 43 min

Go to Digital >

12. Voltage regulation

Note:

1. R_L denotes load resistance.
2. The value of knee current $I_{knee} = 0$



Answer Options

Select any one option

☐ 4kΩ

☐ 5kΩ

☐ 3kΩ

☐ 2kΩ



Submit test

Analog ends in 44 min

Aptitude

Submitted

Analog

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Digital

Starts in 44 min

Go to Digital >

9. An ideal opamp

Which of the given statements is incorrect about an ideal op amp?

Answer Options

Select any one option

☐ Has infinite gain

☐ Has infinite input impedance

☐ Has infinite bandwidth

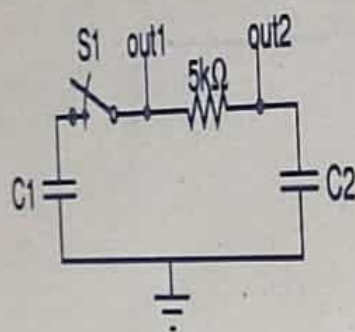
☐ Has infinite output impedance

5. Capacitors C1 and C2

You are working on the following passive circuit. The capacitor values are given as $C1 = 3 \mu\text{F}$ and $C2 = 1 \mu\text{F}$. You have been informed that the capacitor **C1** is precharged to **10V**, while **C2** has no initial charge.

Also, the switch **S1** is initially open and is closed at time, $t = 0$.

What will be the final voltage obtained at the capacitors **C1** and **C2**?



Answer Options

Select any one option

☐ C1 will reach 7.5V, C2 will reach 2.5V

☐ C1 will reach 2.5V, C2 will reach 7.5V

☐ C1 and C2 will both reach 7.5V

☐ C1 will reach 0, C2 will reach 10V

Clear Ans

Submit Test

Time ends in 43 min

Module

Submitted

Analogue

1 2 3

4 5 6

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16 17 18

19 20

Digital

Starts in 43 min

Go to Digital >

12. Voltage regulation

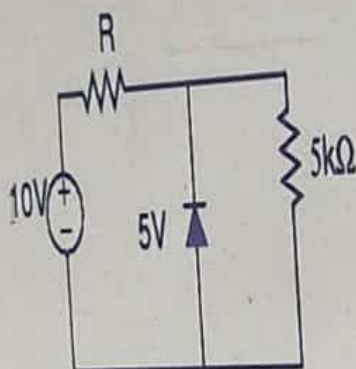
It is given that $V_{in} = 10V$ and $R_L = 5k\Omega$ for the given circuit. You now wish to use the same circuit as a regulator circuit.

What will be the maximum value of R such that a constant voltage is regulated across the load?

Note:

1. R_L denotes load resistance.

2. The value of knee current $I_{knee} = 0$



Answer Options

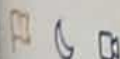
Select any one option

☐ 4kΩ

☐ 5kΩ

☐ 3kΩ

☐ 2kΩ



Submit test

< Previous

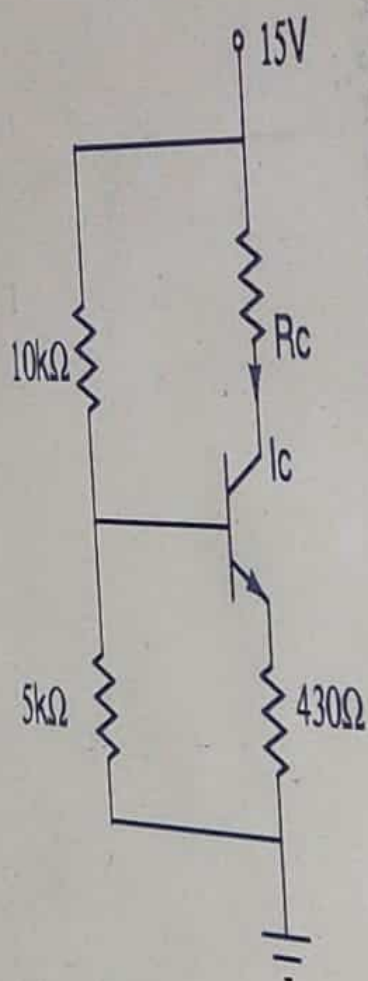
Next >

11. Collector current of a transistor

You have designed the given circuit such that the transistor is operating in the active region. What will be the value of collector current I_c in the given scenario?

Note:

1. The current gain of the transistor, β is very large
2. The Base-emitter voltage of the amplifier is $0.7V$.



Answer Options

< Previous

Next >

6. Modified output of the system

You have designed a linear time-invariant control system that produces an output $c(t)$ for a certain input $r(t)$. You decide to modify $r(t)$ by passing it through a block whose transfer function is e^{-s} .

If the modified input is applied to the system, what will be the modified output of the system?

Note:

1. In the given transfer function e^{-s} , e denotes an exponential function and s denotes the Laplace operator.

Answer Options

Clear Answer

Select any one option

☐ $c(t-1)$

☐ $c(t) \cdot u(t-1)$

☐ $c(t) / 1+e^{-1}$

☐ $c(t) / 1+e^{-s}$

< Previous

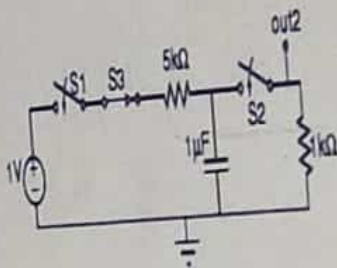
Next >

19. A passive network

A passive network is connected, as shown in figure. It is observed that the switch **S1** is initially open, and is closed at time $t = 0$. The switch **S2** is also initially open and is closed at **3 ms**. However, the switch **S3** is initially closed and is opened only at time **2.5 ms**.

At what time will **out2** reach **0.2V**?

Note: Assume that capacitors have no initial charge.



Answer Options

Select any one option

Clear Answer

☐ $t = 2.67\text{ms}$

☐ $t = 3.67\text{ms}$

☐ $t = 4.67\text{ms}$

☐ $t = 5.12\text{ms}$

Submit Test

in 43 min

13. Series RLC circuit

You have a series RLC circuit with resistance $10\ \Omega$, inductance 1 mH , and capacitance $10\ \mu\text{F}$. Assume that a sinusoidal input of the form $V \sin(\omega t)$ is applied to the circuit.

In the given context, for what value of ω will the current flowing in the circuit be in the form $I \sin(\omega t)$?

Answer Options

Select any one option

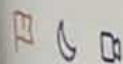
☐ 10M rad/s

☐ 100M rad/s

☐ $100/(2\pi)\text{ K rad/s}$

☐ 10 K rad/s

Clear All



Submit test

Analog ends in 43 min

Aptitude

Submitted

Analog

- 1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

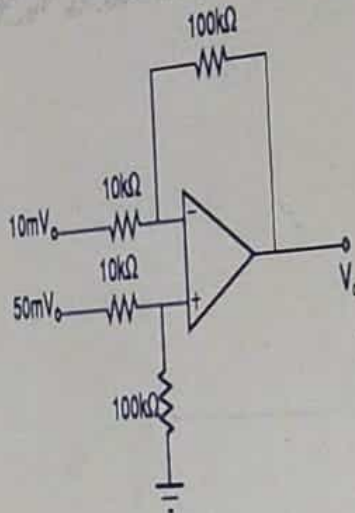
Digital

Starts in 43 min

Go to Digital >

20. An operational amplifier

In the circuit shown below, the operational amplifier is ideal in nature. Find the output voltage V_o .



Answer Options

Select any one option

☐ 500mV

☐ 100mV

☐ 600mV

☐ 400mV



Submit test

< Previous

Next >

15. Rms and average values

It is given that x_r and x_a are respectively the rms and average values of $x(t) = x(t-T)$. Similarly, y_r and y_a are respectively the rms and average values of $y(t) = kx(t)$.

Which of the following equations hold **TRUE** in the given scenario?

Note: It is given that k and T are independent of t .

Answer Options

Select any one option

☐ $y_a \neq kx_a : y_r = kx_r$

☐ $y_a = kx_a : y_r \neq kx_r$

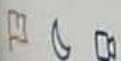
☐ $y_a \neq kx_a : y_r \neq kx_r$

☐ $y_a = kx_a : y_r = kx_r$

Digital

Starts in 43 min

Go to Digital >

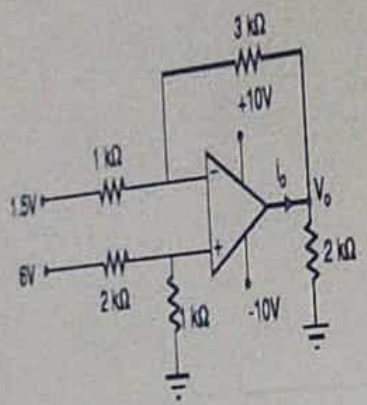


Submit test

Analogy ends in 43 min

16. An ideal opamp circuit

Find the value of the output voltage V_o for the given ideal op amp circuit.



Answer Options

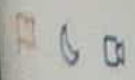
Select any one option

- ☐ 3.5V
- ☐ 4.0V
- ☐ 4.5V
- ☐ 3.0V

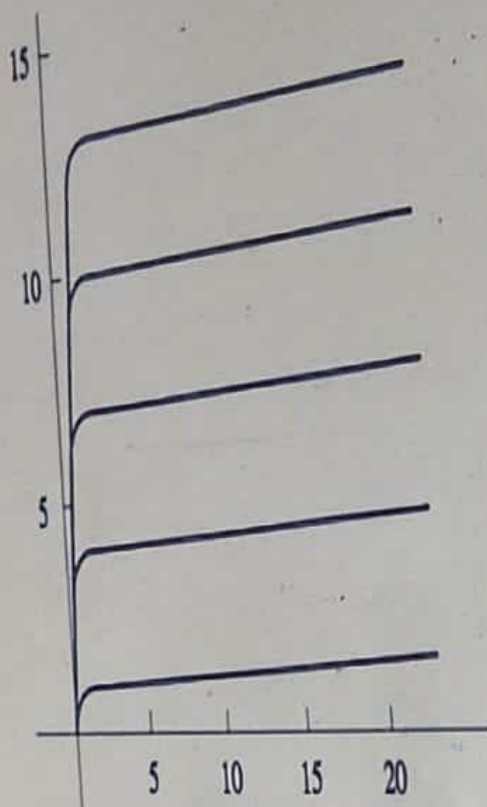
Digital

Starts in 43 min

Go to Digital >



17. Characteristics of a BJT



Answer Options

Select any one option

- ☐ The graph has collector-emitter voltage along x-axis.
- ☐ The graph has collector current along the y-axis.
- ☐ Each curve in the graph corresponds to a different value of base current.
- ☐ None of these

Analog ends in 44 min

Apptude

Submitted

Analog

- 1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 44 min

Go to Digital >

1. The Superposition theorem ✓

You wish to apply Superposition theorem to a network X.

What can X be in the given context?

Answer Options

Select any one option

- ☒ A Linear Bilateral Network
- ☐ A Non-linear Bilateral Network
- ☐ A Unilateral network
- ☐ All of these

< Previous

2. Maximum power

A voltage source $v(t) = V \sin 450\pi t$ Volts having an internal impedance of $(5+j12)\Omega$ is connected to a purely resistive load R_L . What should be the value of R_L (in Ω) such that it can draw the maximum power out of the source?

Answer Options

Select any one option

☐ 2 Ω

☐ 3 Ω

☐ 4 Ω

☐ 13 Ω

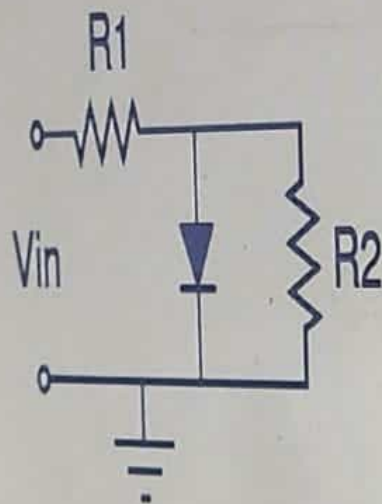
Analog ends in 44 min

4. Minimum value of resistance

The given diode circuit has the following parameters:

1. Input supply voltage, $V_{in} = 4V$
2. Resistance $R_1 = 6.6k\Omega$
3. Cut-in voltage of diode, $V_f = 0.7V$

Calculate the minimum value of resistance R_2 for which the diode will be ON.



Answer Options

Select any one option

☐ 1.4 k Ω

☐ 1.8 k Ω

☐ 2.4 k Ω

☐ 3 k Ω

Analog ends in 44 min

Apptude

Submitted

Analog

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Digital

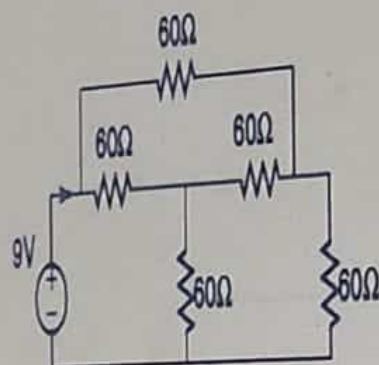
Starts in 44 min

Go to Digital >

3. The input resistance

What is the input resistance as seen from the source side in the given circuit?

Hint: Transform the T-network to a Delta-network.



Answer Options

Select any one option

☐ 270 ohm

☐ 180 ohm

☐ 120 ohm

☐ 60 ohm

Analog ends in 44 min

Apptitude

Submitted

Analog

- 1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 44 min

Go to Digital >

8. Resolution of an ADC

You have a 10-bit ADC with input ranging from 0 to 10V.

What will be the resolution (step-size) of the ADC?

Answer Options

Select any one option

☐ 0.977 mV

☐ 9.77 mV

☐ 97.7 mV

☐ 977 mV

Analog ends in 43 min

Apptude

Submitted

Analog

- 1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 43 min

Go to Digital >

18. A cascade of three linear, time-invariant system

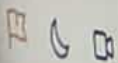
It is given that a cascade of three linear, time-invariant systems is causal and unstable.

Which of these conclusions can be made about the cascaded system?

Answer Options

Select any one option

- ☐ Each system in the cascade is individually causal and unstable.
- ☐ At least one system is unstable and at least one system is causal.
- ☐ At least one system is causal and all systems are unstable.
- ☐ The majority of the systems are unstable and causal.



Submit test

Digital ends in 44 min

4. A bistable multivibrator

Apptude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

You want to design a bistable multivibrator to increase the frequency at which the data is stored. When doing so, you want to ensure that the data is stored at the rising and falling edge of the clock. However, due to lack of resources, you only have MUXes to implement this.

How many MUXes will you need to achieve your objective?

Answer Options

Select any one option

☐ 2

☐ 3

☐ 4

☐ 5

Clear Answer



Submit Test

Digital ends in 45 min

1. RTL Technology advantage

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

What is the primary advantage of using RTL technology?

Answer Options

Select any one option

☐ It results as low power dissipation

☐ It uses a minimum number of resistors

☐ It uses a minimum number of transistors

☐ It operates swiftly

S MIND JOSEPH

EMAN THINKIN

CIRCUIT DESIGN

Digital ends in 44 min

< Previous

Next >

6. The equivalent function

Apitude

Submitted

Which of these functions is the same as the following function?

$$f = [(x \cdot y) + (x \cdot y)]$$

Analog

Submitted

Digital

1 2 3

4 5 6

Answer Options

7 8 9

Select any one option

10 11 12

☐ $f = (x+y)(x+y)$

Clear Answer

13 14 15

☐ $f = [(x+y)(x+y)]$

16 17 18

☐ $f = (x \cdot y)(x \cdot y)$

19 20

☐ None of these



Submit Test

< Previous

Next >

Digital ends in 44 min

7. Code synthesis

Apptitude

Submitted

Analogy

Submitted

Digital

```
wire [255:0] a;  
wire [255:0] b;  
wire [255:0] c;
```

```
assign c = sel_line ? a : b;
```

Which of the following can be inferred from the above given RTL code snippet?

1. The code on synthesis will translate into a relatively slow design.
2. The sel_line has 256 loads.
3. The code synthesizes into 256 parallel MUX structures.

Answer Options

Select any one option

☐ 1,2

Clear Answer

☐ 1,3

☐ 2,3

☐ 1,2,3



Digital ends in 44 min

8. Branches in a graph

Amplitude

Submitted

Analog

Submit

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

The graph of a network has 10 nodes and 4 independent loops.

Find the number of branches in the graph.

Answer Options

Select any one option

☐ 11

☐ 13

☐ 14

☐ 12

Clear Answer



Submit test

Digital ends in 44 min

10. Executing a RISC instruction set

< Previous Next >

Apptitude

Submitted

Analog

Submitted

Digital

You are executing an instruction set in a RISC processor.

In how many instruction stages can you complete this execution?

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Answer Options

Select any one option

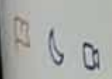
☐ 4

☐ 5

☐ 3

☐ 7

Clear Answer



Submit Test

< Previous

Next >

Digital ends in 45 min

3. A flash memory

Apitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Which of these statements doesn't describe Flash memory?

Answer Options

Select any one option

Clear Answer

☐ It can erase data byte-wise.

☐ It is used to rewrite the data frequently.

☐ NOR gates are used to design this memory.

☐ Both Choice 1 and 3

☐ None of these



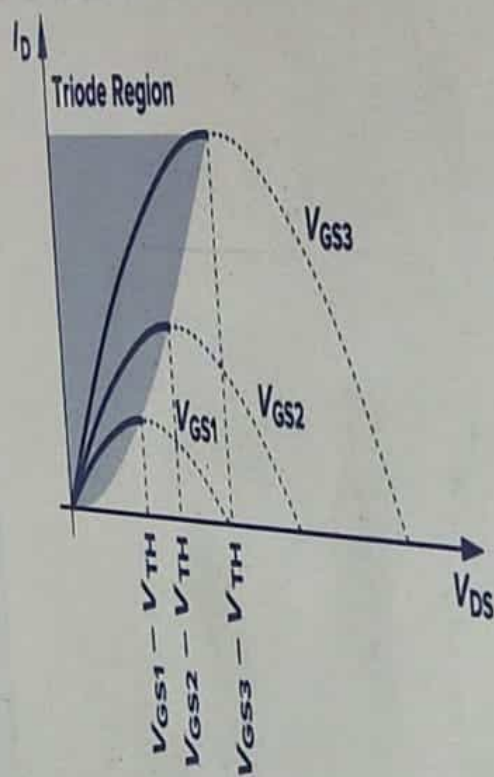
Submit test

Digital ends in 44 min

11. The IV characteristics of an nMOS

The IV characteristics of an nMOS, as predicted by the standard equation of drain current I_D , are given in the following graph.

In reality, the drain current does not follow the parabolic behavior and I_D becomes relatively constant, and we say the device operates in the "saturation" region. What could be a valid reason behind this?



Answer Options

Select any one option.

☐ Some channel still exists even when $V_{DS} > V_{GS} - V_{TH}$

☐ Channel is pinched off but current flows due to minority charge carriers

☐ Upon passing the pinch-off point, the electrons simply shoot through the depletion region near the drain junction

Clear Answer

< Previous

Next >

Time ends in 45 min

2. No. of cache sets

You are working on a 16-bit microprocessor that uses a 2-way set-associative cache with a total capacity of 256 bytes. It is given that each cache line contains 8 bytes.

How many cache sets are there in the cache?

1 2 3

4 5 6

Answer Options

7 8 9

Select any one option

10 11 12

☐ 8

Clear Answer

13 14 15

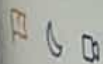
☐ 16

16 17 18

☐ 32

19 20

☐ 64



Submit Test

< Previous

Next >

Digital version 44 min

9. Conditioning in Latch

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8

10 11 12

13 14 15

16 17 18

19 20

Assume that you have designed a latch using NAND and NOR Gates.

Which condition enables this configuration to remain in the latched condition?

Answer Options

Select any one option

☐ Low input voltages

☐ Synchronous operation

☐ Gate impedance

☐ Cross-Coupling

Clear Answer

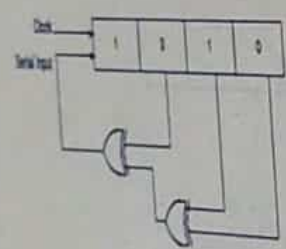


Digital ends in 44 min

5. Working on a shift register

You are working on the given shift register. You have initially loaded it with the bit pattern **1010**. Subsequently, the shift register is clocked. On clocking, it is observed that with each clock pulse the pattern gets shifted by one bit position to the right. With each shift, the bit at the serial input is pushed to the left most position (MSB).

How many clock pulses will the shift register take such that its content becomes **1010** again?



Answer Options

Select any one option

☐ 3

☐ 7

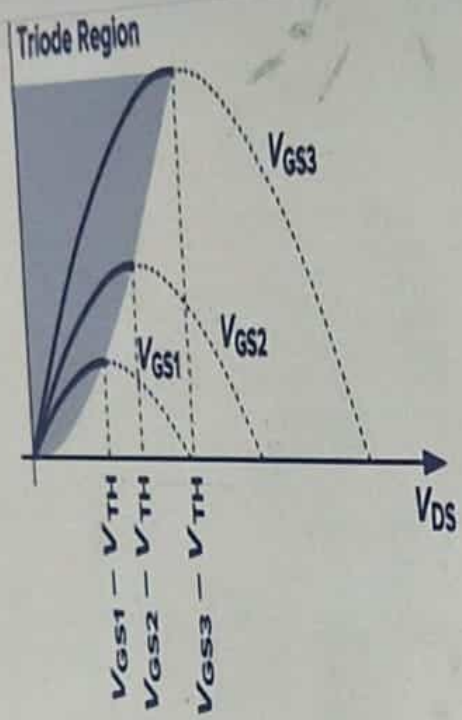
☐ 11

☐ 15

Clear Answer

11. The IV characteristics of an nMOS

11. The IV characteristics of an nMOS



Answer Options

Select any one option

☐ Some channel still exists even when $V_{DS} > V_{GS} - V_{TH}$

Clear Answer

☐ Channel is pinched off but current flows due to minority charge carriers

☐ Upon passing the pinch-off point, the electrons simply shoot through the depletion region near the drain junction and arrive at the drain terminal.

☐ V_{DS} is so high that it causes the current flow from the bulk to the drain terminal via the substrate.

Digital ends in 44 min

12. Response of a system

A control system is defined by the given mathematical relationship:

$$d^2x/dt^2 + 6 \cdot dx/dt + 5x = 12(1 - e^{-2t}), \text{ where } x(t=0) = 0 \text{ and } x'(t=0) = 0$$

Calculate the response of the system as $t \rightarrow \infty$.

< Previous

Next >

Apptude

Control

Analogy

Submit

Digital

1 2 3

4 5 6

Answer Options

7 8 9

Select any one option

10 11 12

Clear Answer

13 14 15

16 17 18

19 20

☐ $x = 6$

☐ $x = 2$

☐ $x = 24$

☐ $x = -2$



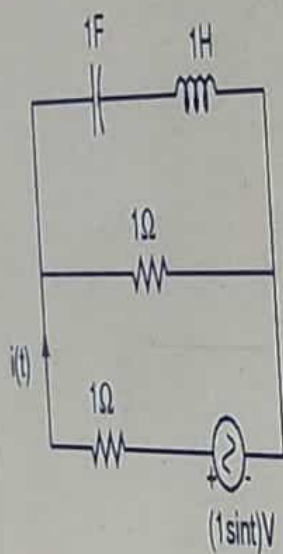
< Previous

Next >

Digital ends in 44 min

13. Finding rms value of current

Find the rms value of the current $i(t)$ in the circuit shown below.



Answer Options

Select any one option

☐ $1/2\text{ A}$

☐ $1/2\text{ A}$

☐ 1 A

☐ 2 A

Clear Answer



Digital ends in 44 min

14. Reducing dynamic power consumption

You want to reduce the dynamic power consumption in a certain system. It is given that the system comprises of a newly developed block which will multiply the data passed into the system at every clock cycle. The steady state of the system is close to 0. However, due to noises, it is expected that the system will be fluctuating about the steady state.

You want to ensure that the dynamic power loss is minimum as the value fluctuates. In which of these forms should numbers be stored in the system so as to achieve the same?

Apptitude

Submitted

Analogy

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Answer Options

Select any one option

Clear Answer

☐ 1s Complement

☐ 2s Complement

☐ Signed Magnitude

☐ Dynamic Power is Independent of Number System Representation



< Previous

Next >

Quiz ends in 44 min

15. Using sizeof()

Predict the output of the following code snippet.

```
class temp{
    int x;
    temp node;
    float y;
};

main()
{
    temp head = new temp();
    display(sizeof(head), sizeof(temp));
}
```

Answer Options

Select any one option

☐ 8 24

☐ 24 24

☐ 4 12

☐ 8 12

Clear Answer

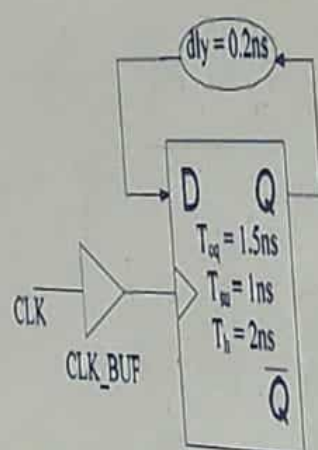
Digital ends in 43 min

16. Type of violation and delay

Analyze the given circuit and choose the option that answers the below question correctly.

- A. What is the type of violation of the circuit?
- B. What is the delay of CLK_BUF that will increase the frequency of operation of the circuit by 1.5x?

Note: Assume delay of CLK_BUF is 0.



Answer Options

Select any one option

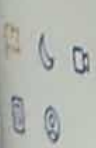
- A - Setup Violation
- B - 0.5ns

Clear Answer

- A - Hold violation
- B - CLK_BUF delay does not affect frequency of operation

- A - Hold Violation
- B - 0.5 ns

- A - No violation
- B - CLK_BUF delay does not affect frequency of operation



Submit Answer

< Previous

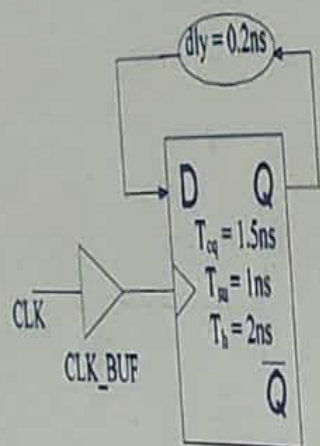
Next >

Digital ends in 43 min

16. Type of violation and delay

Q. What is the delay of CLK_BUF that will increase the frequency of operation of the circuit by 10%?

Note: Assume delay of CLK_BUF is 0.



Answer Options

Select any one option

☐ A - Setup violation
B - 0.5ns

Clear Answer

☐ A - Hold violation
B - CLK_BUF delay does not affect frequency of operation

☐ A - Hold Violation
B - 0.5 ns

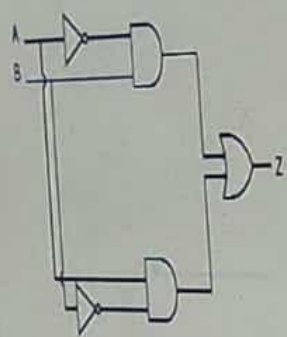
☐ A - No violation
B - CLK_BUF delay does not affect frequency of operation

Submit test

Digital logic in 45 min

17. The correct equivalent circuit

In the circuit shown below, A and B are the digital inputs and Y is the digital output.
Identify the correct equivalent circuit for the same.



- 1 2 3
- 4 5 6
- 7 8 9
- 10 11 12
- 13 14 15
- 16 17 18
- 19 20

Answer Options

Select any one option

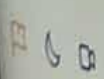
☐ XOR gate

☐ XNOR gate

☐ NAND gate

☐ NOR gate

Clear



Submit Test

< Previous

Next >

Digital ends in 43 min

18. Understanding strings in C

Predict the output of the following C code snippet.

```
#include <stdio.h>
int main() {
    char ABC[5] = "hello";
    printf("%d", *(ABC + 4));
    return 0;
}
```

Answer Options

Select any one option

☐ 111

☐ 0

☐ Garbage value

☐ Runtime error

Clear Ans



Submit Test

< Previous

Next >

Digital ends in 43 min

20. A VHDL code

Analyse the given VHDL code and determine what is being implementing using it.

Apptude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19

```
entity example is
    port(
        Q : out std_logic;
        CLK : in std_logic;
        CE : in std_logic;
        RESET : in std_logic;
        D : in std_logic;
        SET : in std_logic
    );
end example;
```

```
architecture Behavioral of example is
begin
    process(CLK)
    begin
        if (rising_edge(CLK)) then
            if (RESET = '1') then
                Q <= '0';
            else
                if (SET = '1') then
                    Q <= '1';
                else
                    if (CE = '1') then
                        Q <= D;
                    end if;
                end if;
            end if;
        end process;
    end Behavioral;
```

Answer Options

Select any one option

Submit test

A behavioural model of a synchronous D flip flop

Digital ends in 43 min

19. Reading values from a DRAM Memory

< Previous

Next >

Apptitude

Submitted

Analogy

Submitted

Digital

The following steps need to be carried out while reading values from a DRAM Memory.

1. Precharging the column line parasitic capacitance to half-threshold.
2. Asserting a single row line.
3. Changing of the bit line voltage slightly above or below threshold depending on whether the value stored is 0 or 1.
4. Reading and amplification of the signal on the column line.

Which of these steps will be impacted directly if the storage capacitor used in a DRAM cell is small?

1 2 3

4 5 6

Answer Options

7 8 9

Select any one option

10 11 12

☐ 1

13 14 15

☐ 2

16 17 18

☐ 3

19 20

☐ 4

Clear Answer



Digital ends in 43 min

20. A VHDL code

Apitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

architecture Behavioral of example is

```
begin
  process(CLK)
  begin
    if (rising_edge(CLK)) then
      if (RESET = '1') then
        Q <= '0';
      else
        if (SET = '1') then
          Q <= '1';
        else
          if (CE = '1') then
            Q <= D;
          end if;
        end if;
      end if;
    end if;
  end process;
end Behavioral;
```

Answer Options

Select any one option:

- ☐ A behavioural model of a synchronous D flip flop
- ☐ A behavioural model of an asynchronous D flip flop
- ☐ A 4 bit synchronous UP counter
- ☐ None of these



Submit test



Technical

1 → A

2 → D

3 → D

4 → A.

5

6 → A

7 → B

8 → B

9 → D.

10 → D/A

11 → C.

12 → B.

13 → D

14 → D (Nearly 5mV)

15 → A

16 → A

17 → D [None of these]

18 → A

19

20 → D (400mV)